Tactix: Competition-Winner Proof-Rule Audit Checkpoint

Report to Andrew

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Outcome

The frozen AM402 source audit gives a qualified yes to the missing-proof-rules question. The verified SC26 LymphoSAT bundle has listed specialized solvers whose UNSAT paths emit general pseudo-Boolean (PB) polynomial arithmetic, redundancy and subproof derivations. Its composite verifies a returned UNSAT certificate with VeriPB before publishing it. The pinned Satsuma source also has PB dominance/arithmetic output. Tactix can emit PB arithmetic in Lucky 19/20, but only for complete original-input terminal refutations at its startup proof boundary. It has no general live PB derivation action or event interface. The separate clausal vocabulary—RUP, RAT, finite substitution redundancy and deletion—appears represented; this source pass did not establish a missing clausal primitive.

Evidence and interpretation

The LymphoSAT `order.json` and generated matcher registry include equivalence-chain, XOR-chain, Tseitin-formula and random-XOR solvers. Their source emits `pol` addition/scaling/division, `red` constraints and/or PB subproofs; the AM402 technical report gives exact source locations. A literal-source count finds `pol` in at least 15 family producers, `red` in at least seven and `pbc` in at least three; these are lower bounds, not measured competition usage. Tactix's Lucky parity and count emitters use some of the same PB commands, so the gap is general applicability and proof-stream integration, not total absence of the arithmetic syntax.

This is a proof-calculus and proof-efficiency design gap, not automatically an N1 failure for the currently declared action catalog. Nor does PB dominance syntax alone prove a missing semantic symmetry action: Tactix's X7 can carry finite clausal substitution witnesses. A general fix would require a typed PB derivation interface, legality and resource bounds, original-CNF proof checking, and a deliberate choice about the present startup-PBP versus ordinary-DSR boundary. Splicing PB commands into a published DSR stream would not satisfy the current N2 contract. The remaining CaDiCaL/Kissat producer and public reachability audits still need to be completed.

Current gates and next action

AM400 and AM401 full Debug jobs `1720786` and `1720898` were running during this audit, and both independent affected Opus owners lacked terminal verdicts. The canonical no-option DSR checker issue H2 and the post-finalization public-addition SAT-model behavior remain open. Therefore no combined N1/N2 pass is claimed; N3/N4 and Rank/B+C/WV performance gates also remain open. Online learning is still prototype-only: its root-pass dosing interface exists, but production use needs default-exact inprocessing slot control and safe-point strategy ownership. Next consume the running correctness evidence, repair any bounded handoff generally, decide the two open public/checker contracts, then complete the source-to-action audit and combined gates. The requested 5,000-second, 4-CPU, 32-GiB full Tier-D hwulab run remains conditional on achieving the goal.

Primary evidence: AM402 technical report; AM398 winner proof-emitter report; pinned sources under `winners/`; `docs/calculus.md` and `DEVELOPMENT_PLAN.md` §4–5; current `PROGRESS.md`.